# Experiment No-22 : Bird Can Fly or Not

## Aim:

To write a Prolog program to check whether a particular bird can fly or not, and to incorporate the required queries.

## Programming Code:

```
% ---------- Facts ----------
bird(sparrow).
bird(eagle).
bird(pigeon).
bird(parrot).
bird(crow).
bird(penguin).
bird(ostrich).
bird(emu).
bird(kiwi).

% Birds that cannot fly (exceptions)
cannot_fly(penguin).
cannot_fly(ostrich).
cannot_fly(emu).
cannot_fly(kiwi).

% ---------- Rules ----------
% A bird can fly if it is a bird and is not an exception
can_fly(X) :- bird(X), \+ cannot_fly(X).

% Print the result for a particular bird
check_bird(X) :-
    (   \+ bird(X)
    ->  format("~w is not in the bird database.~n", [X])
    ;   can_fly(X)
    ->  format("~w can fly.~n", [X])
    ;   format("~w cannot fly.~n", [X])
    ).
```

**OUTPUT:**

```
$ swipl -q exp22.pl

?- can_fly(eagle).
true.

?- can_fly(penguin).
false.

?- can_fly(B).
B = sparrow ;
B = eagle ;
B = pigeon ;
B = parrot ;
B = crow.

?- cannot_fly(B).
B = penguin ;
B = ostrich ;
B = emu ;
B = kiwi.

?- check_bird(parrot).
parrot can fly.
true.

?- check_bird(ostrich).
ostrich cannot fly.
true.

?- check_bird(dog).
dog is not in the bird database.
true.

$
```

## Result

The program to check whether a particular bird can fly or not was implemented successfully using Prolog. Birds listed as exceptions (penguin, ostrich, emu, kiwi) are reported as unable to fly, and all other birds in the database are reported as able to fly.